# Diffusion Temporal Dynamics I

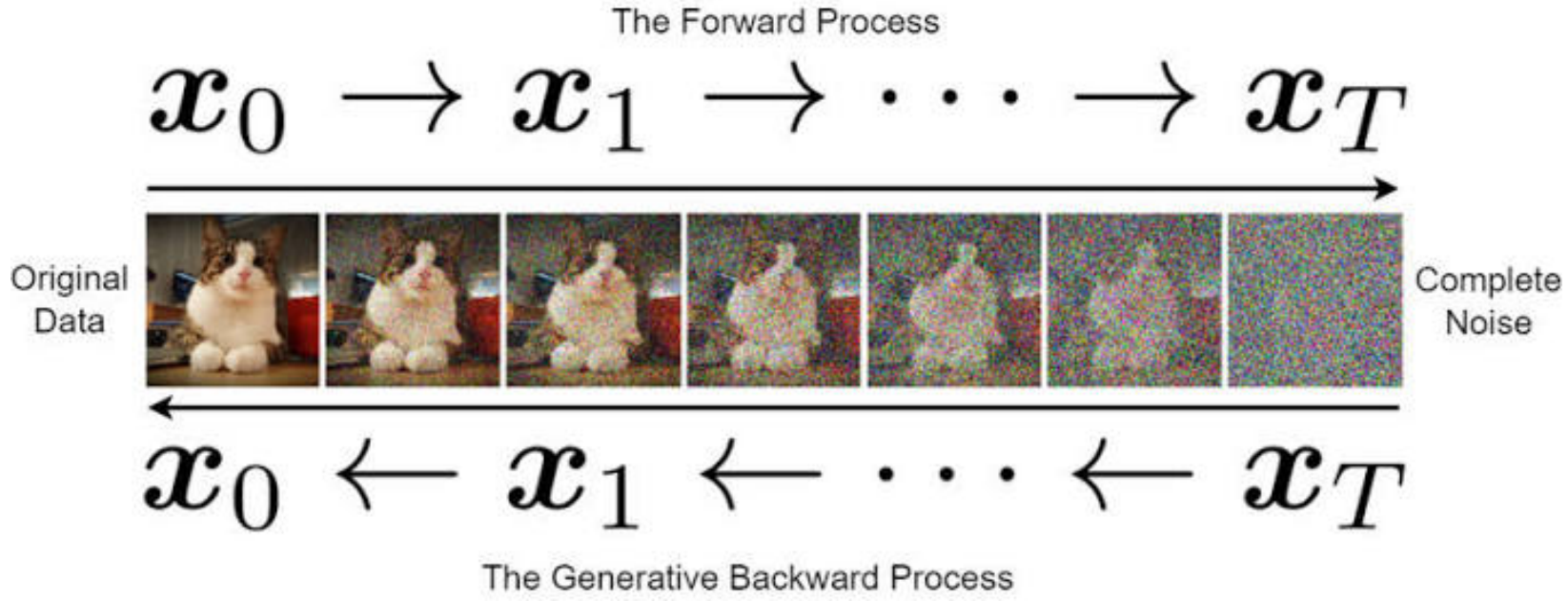
## Denoising Diffusion Probabilistic Models (DDPM), Conditional Diffusion, DDIM

Niruhan Viswarupan

IDSL
Western University

2026-10-11

# Diffusion Models: The Basic Idea

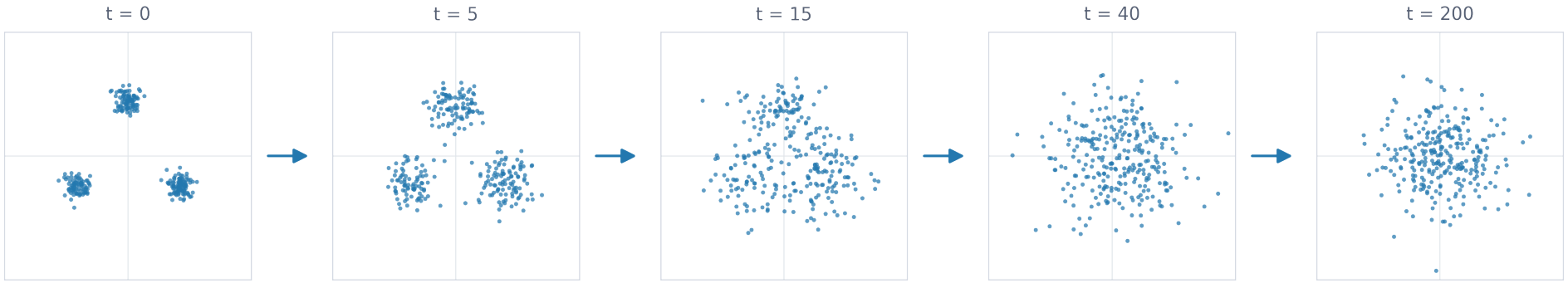



Source: Ghojogh, B., & Ghodsi, A. (2024). *Diffusion Models: Tutorial and Survey.* doi:10.31219/osf.io/w7jcm

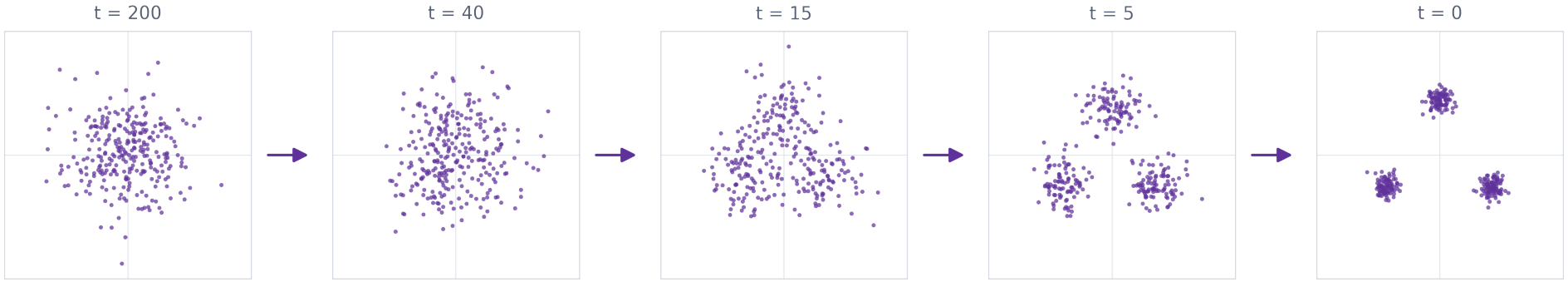
**Forward:** Gradually corrupt a clean image with noise.
**Training:** Teach a neural network to predict the noise in an image
**Reverse:** Use a trained network to progressively transform noise into an image.

# Diffusion: From Structure to Noise and Back

**Forward:** Brownian noise + a weak drift toward the origin.



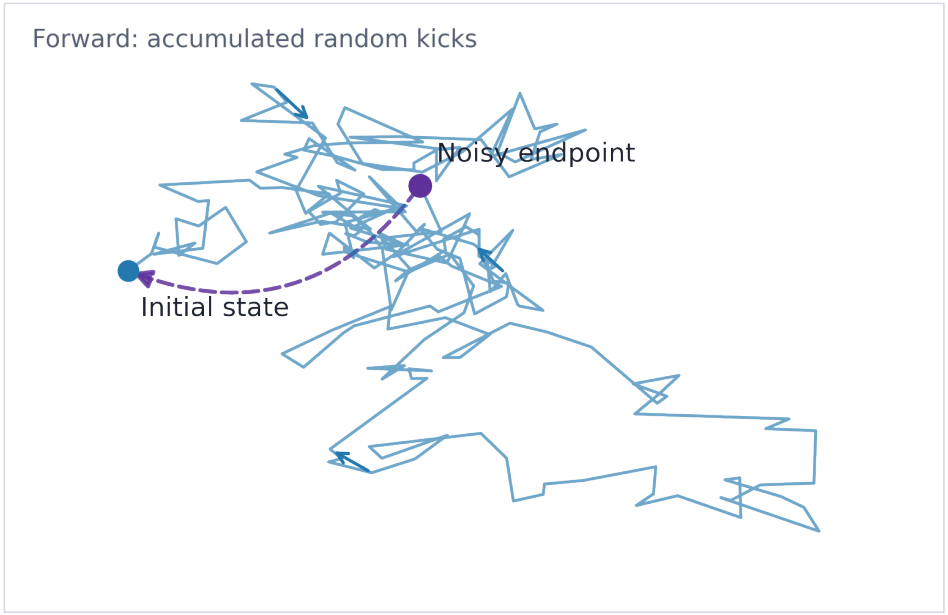

**Reverse:** stochastic steps with a learned, structure-restoring drift.




Each point is a 2D state. Reverse trajectories are independent; the distribution is restored, not the original paths.

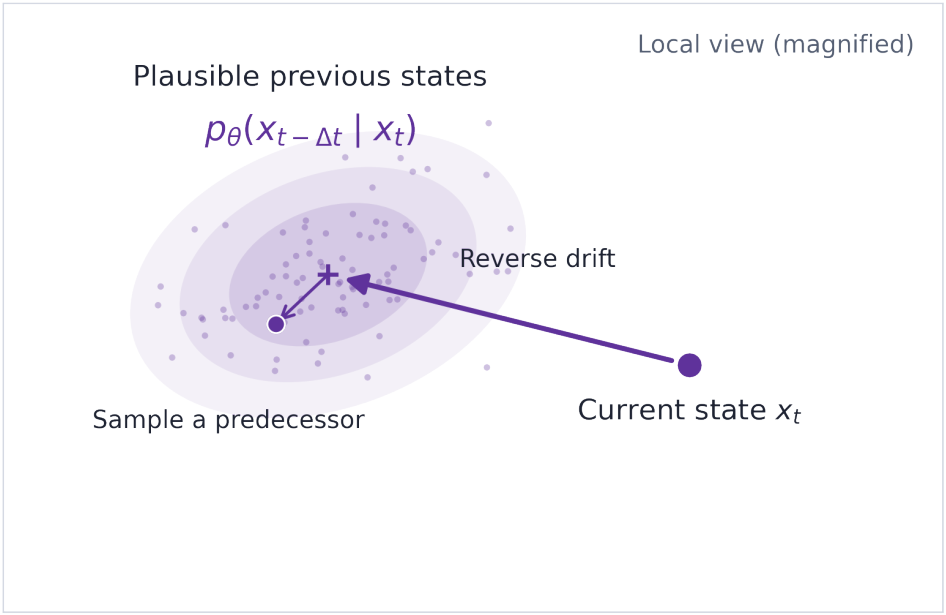
# Why Do We Take Small Steps?

## One large reverse jump




The noisy endpoint does not determine the past. A long reverse transition can have a complex distribution.

## One small reverse step




For sufficiently small steps, the reverse conditional is approximately Gaussian: learn its mean and covariance.

**Many simple reverse steps can generate a complex distribution.**

# Images as Points in a High-Dimensional Space

Represent a clean image as a vector of pixel values:

$$x_0 \in \mathbb{R}^d, \quad d = H \times W \times C.$$

- $H, W$: image height and width.
- $C$: number of channels, with $C = 3$ for RGB.
- $x_0$: the clean image, before adding noise.

Training images are samples from an **unknown data distribution**:

$$x_0 \sim q_{\text{data}}(x_0).$$

> **Learning goal:** Learn a sampling procedure whose generated images follow a distribution close to $q_{\text{data}}$.

# DDPM: Forward Process

# The Forward Diffusion Process

Let $x_0 \in \mathbb{R}^d$ be a clean image sampled from $q_{\text{data}}(x_0)$.

The forward process is a **Markov chain**: each step depends only on the previous state.

$$q(x_{1:T} \mid x_0) = \prod_{t=1}^{T} q(x_t \mid x_{t-1}).$$

### Gaussian transition

$$q(x_t \mid x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t} x_{t-1}, \beta_t I).$$

Here, $\alpha_t = 1 - \beta_t$
and $0 < \beta_t < 1$.

### Equivalent sampling rule

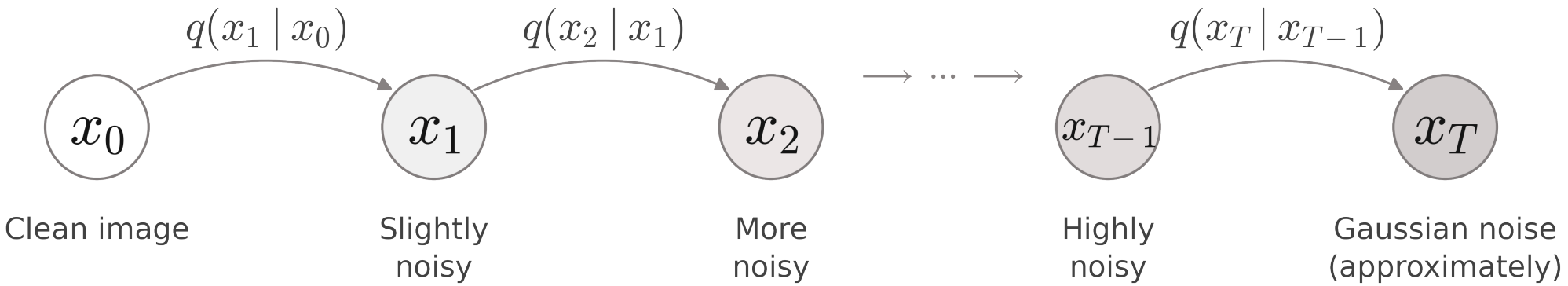
$$x_t = \sqrt{\alpha_t} x_{t-1} + \sqrt{\beta_t} z_t.$$

$z_t \sim \mathcal{N}(0, I)$, independently across steps and independent of $x_0$.

> At each step, **shrink the previous image** and **add fresh Gaussian noise**.

# Markov Diagram

FORWARD → Gradually add Gaussian noise




**DDPM Forward Process as Markov chain.** The forward process progressively corrupts clean data with Gaussian noise.

# Sampling Any Noisy State Directly

Define the **cumulative signal retention**:

$$\overline{\alpha}_t = \prod_{s=1}^{t} \alpha_s.$$

Composing the Gaussian transitions gives:

$$q(x_t \mid x_0) = \mathcal{N}\Big(x_t; \sqrt{\overline{\alpha}_t}x_0, (1-\overline{\alpha}_t)I\Big).$$

**Why a product? Compose two steps.**
With independent $z_1, z_2 \sim \mathcal{N}(0, I)$:

$$\begin{aligned} x_1 &= \sqrt{\alpha_1}x_0 + \sqrt{\beta_1}z_1, \\ x_2 &= \sqrt{\alpha_2}x_1 + \sqrt{\beta_2}z_2 \\ &= \sqrt{\alpha_2\alpha_1}x_0 \\ &\quad + \sqrt{\alpha_2\beta_1}z_1 + \sqrt{\beta_2}z_2. \end{aligned}$$

The combined noise is Gaussian with variance:

$$\alpha_2\beta_1 + \beta_2 = 1 - \alpha_2\alpha_1.$$

Repeating gives signal coefficient $\sqrt{\overline{\alpha}_t}$ and noise covariance $(1-\overline{\alpha}_t)I$.

## Equivalent sampling rule

$$x_t = \sqrt{\overline{\alpha}_t}x_0 + \sqrt{1-\overline{\alpha}_t}\varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I).$$

We can sample $x_t$ **directly from a clean image**, without simulating the preceding $t$ steps.

# Conditional and Marginal Distributions

Across all clean images, the noisy-image distribution is

$$q_t(x_t) = \int q(x_t \mid x_0) q_{\text{data}}(x_0)\, \mathrm{d}x_0.$$

**Conditional: one clean image**

$$q(x_t \mid x_0)$$

For fixed $x_0$, a Gaussian centered at $\sqrt{\overline{\alpha}_t} x_0$.

**Marginal: all clean images**

$$q_t(x_t)$$

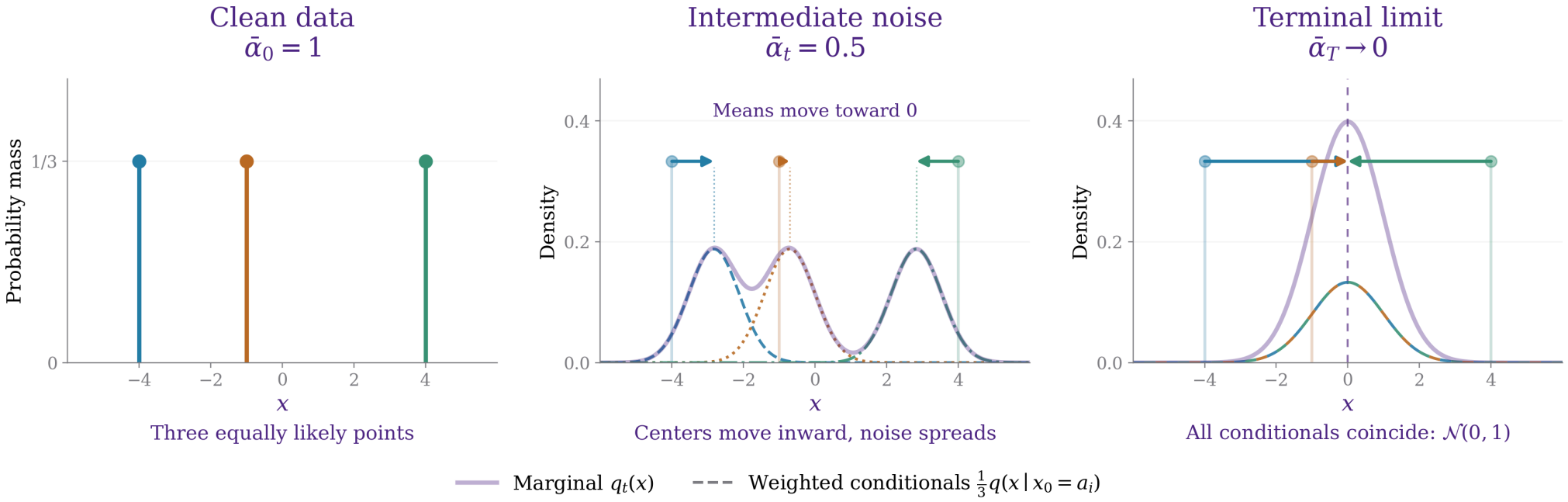
A mixture of Gaussians weighted by $q_{\text{data}}(x_0)$, generally non-Gaussian.

When $\overline{\alpha}_T \approx 0$, the influence of the clean image becomes negligible:

$$q_T(x_T) \approx \mathcal{N}(x_T; 0, I).$$

# Forward Diffusion in One Dimension






Each clean point becomes a Gaussian with mean $\sqrt{\overline{\alpha}_t} a_i$ and variance $1 - \overline{\alpha}_t$.

As $\overline{\alpha}_t \to 0$, all components approach $\mathcal{N}(0, 1)$, and so does their mixture.

# DDPM: Reverse Process

# Bayes' Rule

**Bayes' rule** lets us reverse a conditional probability:

$$p(A \mid B) = \frac{p(B \mid A)p(A)}{p(B)}.$$

It follows by writing the same **joint probability** in two ways:

$$\begin{aligned} p(A, B) &= p(A \mid B)p(B) \\ &= p(B \mid A)p(A). \end{aligned}$$

**Posterior** $p(A \mid B)$
How plausible $A$ is after observing $B$.

**Likelihood** $p(B \mid A)$
How plausible the observation $B$ is if $A$ holds.

**Prior** $p(A)$
How plausible $A$ is before observing $B$.

**Evidence** $p(B)$
The overall probability of observing $B$. It normalizes the posterior so probabilities sum to one.

# Bayes' Rule: Example

**Hypothetical example:** 1% have a disease. The test has **99% sensitivity** and **95% specificity** (5% false positives).

**You test positive. What is the probability you have the disease?**

**Using Bayes' rule**

Let $D$ mean disease and $D^c$ mean healthy.

$$P(D \mid +) = \frac{P(+ \mid D)P(D)}{P(+ \mid D)P(D) + P(+ \mid D^c)P(D^c)}.$$

$$= \frac{0.99 \times 0.01}{0.99 \times 0.01 + 0.05 \times 0.99} \approx 16.7\%.$$

**Consider 10,000 people**

| | **Positive** | **Negative** | **Total** |
|---|---|---|---|
| Disease | 99 | 1 | 100 |
| Healthy | 495 | 9,405 | 9,900 |
| **Total** | **594** | **9,406** | **10,000** |

**Why so low?** The disease is rare. The much larger healthy population produces **495 false positives**, compared with **99 true positives**.

$$\underbrace{P(+ \mid D) = 99\%}_{\text{Positive given disease}} \neq \underbrace{P(D \mid +) \approx 16.7\%}_{\text{Disease given positive}}.$$

The positive test still raises the probability from **1% to about 17%**

# Bayes' Rule in Reverse Diffusion

The same rule applies to **probability densities**:

$$\underbrace{q(x_{t-1} \mid x_t)}_{\text{reverse transition}} = \frac{\overbrace{q(x_t \mid x_{t-1})}^{\text{forward transition}} \overbrace{q_{t-1}(x_{t-1})}^{\text{prior density}}}{\underbrace{q_t(x_t)}_{\text{normalizing density}}}.$$

Given a noisy state $x_t$, we ask:
**Which previous state $x_{t-1}$ could have produced it?**

A candidate previous state is likely when it:

- Could plausibly **produce the observed noisy state** (forward transition).
- Is itself **plausible at time** $t-1$ (prior density).

# What Must the Reverse Process Learn?

$$q(x_{t-1} \mid x_t) = \frac{q(x_t \mid x_{t-1}) q_{t-1}(x_{t-1})}{q_t(x_t)}.$$

The noisy data densities $q_{t-1}$ and $q_t$ are **unknown**, so we cannot evaluate this directly.

Instead, define a **generative model**:

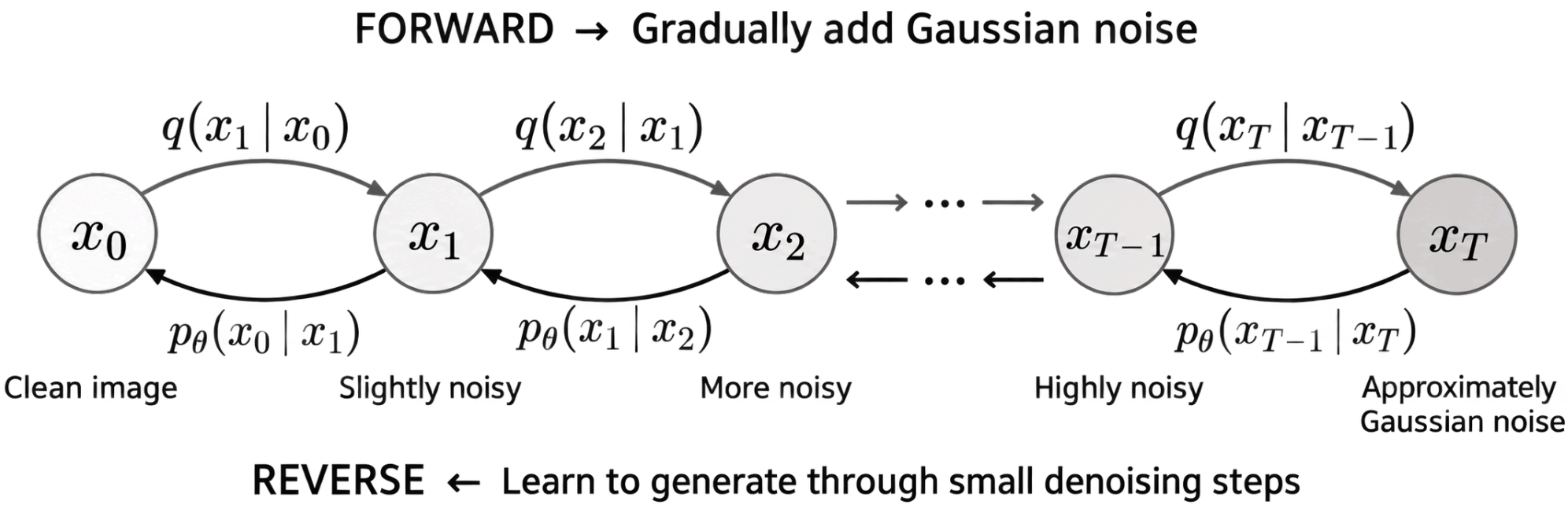
$$p_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t), \quad p(x_T) = \mathcal{N}(x_T; 0, I).$$

**Learned Gaussian reverse transition**

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \sigma_t^2 I).$$

The network learns the **reverse mean** $\mu_\theta(x_t, t)$.

# Markov Diagram




**DDPM as two Markov chains.** The forward process progressively corrupts clean data with Gaussian noise. The learned reverse process generates data through successive denoising steps. Each transition depends only on the current state.

# The Reverse Posterior Given the Clean Image

The true reverse transition need not be Gaussian. A Gaussian model is appropriate for **sufficiently small steps**.

Conditioning additionally on the clean image gives an **exact Gaussian posterior** for $t \geq 2$:

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}\left(x_{t-1}; \tilde{\mu}_t(x_t, x_0), \tilde{\beta}_t I\right).$$

Its mean and variance are available analytically:

$$\tilde{\mu}_t(x_t, x_0) = \frac{\sqrt{\overline{\alpha}_{t-1}}\beta_t}{1-\overline{\alpha}_t} x_0 + \frac{\sqrt{\alpha_t}(1-\overline{\alpha}_{t-1})}{1-\overline{\alpha}_t} x_t.$$

$$\tilde{\beta}_t = \frac{1-\overline{\alpha}_{t-1}}{1-\overline{\alpha}_t}\beta_t.$$

Refer Appendix IV for the derivation of $\tilde{\mu}_t(x_t, x_0)$

During training, $x_0$ is **known**, so this posterior provides a **tractable training target**.

# Training Through the ELBO

**Maximum likelihood** seeks parameters that assign high probability to the training images.

Equivalently, maximize the expected **log-likelihood**:

$$\mathbb{E}_{q_{\text{data}}}[\log p_\theta(x_0)].$$

Evaluating this objective directly requires integrating over all intermediate states.

Instead, maximize the **evidence lower bound (ELBO)**:

$$\mathcal{J}_{\text{ELBO}} \leq \mathbb{E}_{q_{\text{data}}}[\log p_\theta(x_0)].$$

Here, $\mathcal{J}_{\text{ELBO}}$ denotes the **data-averaged ELBO**, computed using the forward process and learned reverse steps.

Refer Appendix II for why ELBO is a lower bound to the data log likelihood

# The ELBO Training Objective

Average over clean images and their forward trajectories: $q_{\text{data}}(x_0)q(x_{1:T} \mid x_0)$.

Refer Appendix I to learn about KL Divergence

$$\mathcal{J}_{\text{ELBO}} = \mathbb{E}_q \Bigg[ \log p_\theta(x_0 \mid x_1) - D_{\text{KL}}(q(x_T \mid x_0) \parallel p(x_T)) - \sum_{t=2}^{T} D_{\text{KL}}(q(x_{t-1} \mid x_t, x_0) \parallel p_\theta(x_{t-1} \mid x_t)) \Bigg].$$

- **Reconstruction:** Increase the probability assigned to the clean image in the final denoising step.
- **Terminal prior:** Penalize mismatch with the Gaussian prior; constant in $\theta$ for a fixed forward process.
- **Reverse transitions:** Reduce mismatch between learned steps and the exact training posteriors.

With fixed Gaussian variances, maximizing the ELBO's reverse-transition terms is equivalent to minimizing **weighted squared errors between means**.

# Parameterizing the Reverse Mean with Noise

Rearrange the forward sampling equation:

$$x_0 = \frac{x_t - \sqrt{1-\overline{\alpha}_t}\varepsilon}{\sqrt{\overline{\alpha}_t}}.$$

Substituting into the **exact posterior mean** gives:

$$\tilde{\mu}_t(x_t, x_0) = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\overline{\alpha}_t}}\varepsilon\right).$$

Replace the unknown noise with a **neural network prediction**:

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\overline{\alpha}_t}}\varepsilon_\theta(x_t, t)\right).$$

**Predicting the noise determines the reverse mean.**

The network receives $x_t$ and timestep $t$.

# Noise Prediction: Training and Generation

**Predicting noise determines the reverse mean.** $\mathcal{J}_{\text{ELBO}}$ reduces to $\mathcal{L}_{\text{simple}}$ as our distributions $q(x_{t-1} \mid x_t, x_0), p_\theta(x_{t-1} \mid x_t)$ are Gaussian:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{x_0,t,\varepsilon}\left[\|\varepsilon - \varepsilon_\theta(x_t, t)\|^2\right].$$

Refer Appendix III for why ELBO reduces to $\mathcal{L}_{\text{simple}}$

Now we have a recipe for tractable training and sampling!

**Training:** Sample $x_0 \sim q_{\text{data}}$, $t$ uniformly from $\{1, \ldots, T\}$, and $\varepsilon \sim \mathcal{N}(0, I)$. Construct:

$$x_t = \sqrt{\overline{\alpha}_t} x_0 + \sqrt{1 - \overline{\alpha}_t}\varepsilon.$$

**Generation:** Start with $x_T \sim \mathcal{N}(0, I)$. For $t = T, \ldots, 1$, apply:

$$x_{t-1} = \mu_\theta(x_t, t) + \sigma_t z_t.$$

Use fresh $z_t \sim \mathcal{N}(0, I)$ for $t > 1$. Set $z_1 = 0$: **no added noise at the final step**.

# Conditional Diffusion

# Conditional Diffusion

Each image has a **condition** $c$, such as a class label, a text prompt, or another image.

**Goal:** Generate images consistent with that condition.

$$p_\theta(x_0 \mid c) \approx q_{\text{data}}(x_0 \mid c).$$

The **forward corruption** usually remains unchanged:

$$q(x_t \mid x_0, c) = q(x_t \mid x_0).$$

Once the clean image is given, adding noise does not require the condition:

$$x_t = \sqrt{\overline{\alpha}_t} x_0 + \sqrt{1 - \overline{\alpha}_t}\varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I).$$

# Learning the Conditional Reverse Process

The reverse model receives both the **noisy image** and the **condition**:

$$p_\theta(x_{t-1} \mid x_t, c) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t, c), \sigma_t^2 I).$$

Train a noise predictor using paired examples $(x_0, c)$:

$$\mathcal{L}_{\text{cond}} = \mathbb{E}_{(x_0,c),t,\varepsilon}\left[\|\varepsilon - \varepsilon_\theta(x_t, t, c)\|^2\right].$$

- Sample $(x_0, c)$ from the training data.
- Sample $t$ uniformly and $\varepsilon \sim \mathcal{N}(0, I)$.
- Construct $x_t$ and predict the noise **given** $c$.

The condition helps the network decide **which image structures to recover**.

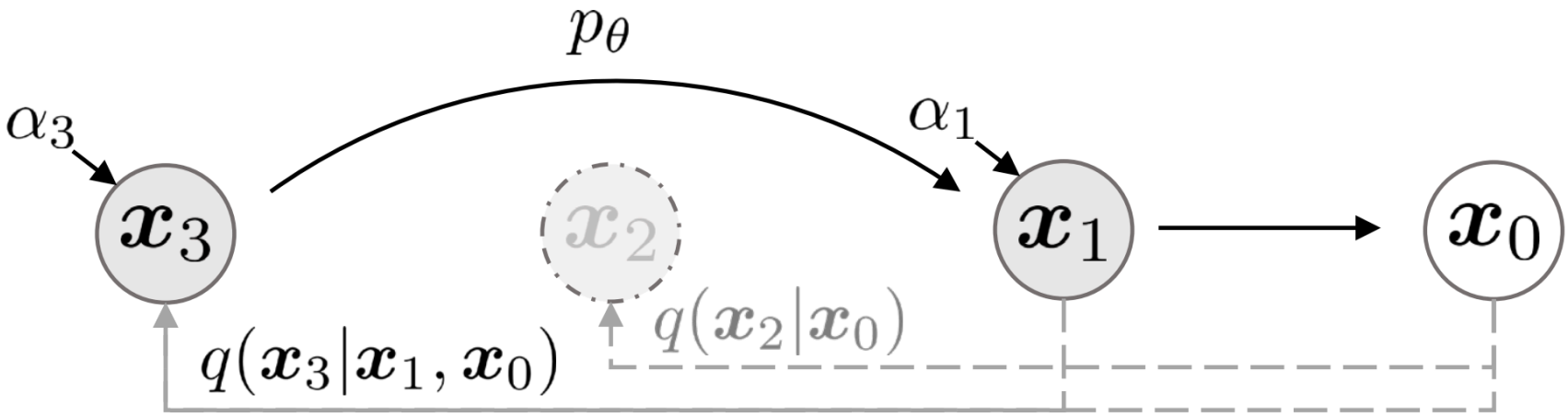
# DDIM

## Denoising Diffusion Implicit Models

# DDIM: The Basic Idea

**Denoising Diffusion Implicit Models (DDIM)** generate images by repeatedly:

1. Estimating the **clean image ($x_0$)** from the current noisy state.
2. Using that estimate to move to a **lower noise level** (e.g: from $x_3$ to $x_1$).




Sampling can skip intermediate timesteps: $x_3 \to x_1 \to x_0$.

# Recover Noise When Clean Image Is Known

The forward marginal is

$$q(x_t \mid x_0) = \mathcal{N}\Big(x_t; \sqrt{\overline{\alpha}_t}x_0, (1-\overline{\alpha}_t)I\Big).$$

Suppose, temporarily, that we **know the clean image** $x_0$. From

$$x_t = \sqrt{\overline{\alpha}_t}x_0 + \sqrt{1-\overline{\alpha}_t}\varepsilon^{(t)},$$

we can recover the noise component:

$$\varepsilon^{(t)} = \frac{x_t - \sqrt{\overline{\alpha}_t}x_0}{\sqrt{1-\overline{\alpha}_t}}.$$

Conditional on $x_0$, this residual follows $\mathcal{N}(0, I)$. The superscript $(t)$ labels the current timestep.

# Construct State at Lower Noise Level

For $s < t$, we want the conditional marginal

$$q(x_s \mid x_0) = \mathcal{N}\Big(x_s; \sqrt{\overline{\alpha}_s}x_0, (1-\overline{\alpha}_s)I\Big).$$

**Estimating the Unknown Quantities**

During generation, the clean image $x_0$ is **unknown**.

Given the current state $x_t$, the network predicts:

$$\hat{\varepsilon}_t = \varepsilon_\theta(x_t, t).$$

Rearranging the noisy-image equation gives:

$$\hat{x}_0^{(t)} = \frac{x_t - \sqrt{1-\overline{\alpha}_t}\hat{\varepsilon}_t}{\sqrt{\overline{\alpha}_t}}.$$

The superscript $(t)$ means **estimated from the current state at time** $t$.

These estimates replace the true clean image and noise in the sampling construction.

# The General Sampling Update

For $s < t$, combine the estimated signal, retained noise direction, and fresh noise:

$$x_s = \underbrace{\sqrt{\overline{\alpha}_s}\hat{x}_0^{(t)}}_{\text{predicted clean signal}} + \underbrace{\sqrt{1-\overline{\alpha}_s-\sigma_{t\to s}^2}\hat{\varepsilon}_t}_{\text{retained noise direction}} + \underbrace{\sigma_{t\to s}z}_{\text{fresh random noise}}.$$

Here, $z \sim \mathcal{N}(0, I)$ is independent of the current state, and $0 \le \sigma_{t\to s}^2 \le 1-\overline{\alpha}_s$.

$\sigma_{t\to s} = 0$ gives the **deterministic DDIM update**.

A positive value introduces **fresh random noise** while reducing the retained-noise coefficient.

Recompute the network prediction at each new state.

# Update in Terms of Network Output

Substitute the clean-image estimate and $\hat{\varepsilon}_t = \varepsilon_\theta(x_t, t)$:

$$x_s = \sqrt{\frac{\overline{\alpha}_s}{\overline{\alpha}_t}}\left(x_t - \sqrt{1-\overline{\alpha}_t}\varepsilon_\theta(x_t,t)\right) + \sqrt{1-\overline{\alpha}_s-\sigma_{t\to s}^2}\varepsilon_\theta(x_t,t) + \sigma_{t\to s} z.$$

**Exact marginal preservation used the true clean image.**

Substituting a network estimate defines an **approximate learned sampler**. Exact marginals are not guaranteed at finite step sizes.

# References & Acknowledgment

**Sohl-Dickstein, J., Weiss, E., Maheswaranathan, N., & Ganguli, S. (2015).**
*Deep Unsupervised Learning using Nonequilibrium Thermodynamics.* ICML.
Proceedings of Machine Learning Research, 37, 2256–2265

**Ho, J., Jain, A., & Abbeel, P. (2020).**
*Denoising Diffusion Probabilistic Models.* NeurIPS.
arXiv:2006.11239

**Dhariwal, P., & Nichol, A. (2021).**
*Diffusion Models Beat GANs on Image Synthesis.* NeurIPS.
arXiv:2105.05233

**Song, J., Meng, C., & Ermon, S. (2021).**
*Denoising Diffusion Implicit Models. ICLR.*
arXiv:2010.02502

Slide content was created with assistance from **ChatGPT Astra**.

# Appendix

# Appendix I: KL Divergence

**KL divergence** measures how distributions differ. In $D_{\mathrm{KL}}(Q \parallel P)$, $Q$ is the **reference** and $P$ its **approximation**.

$$D_{\mathrm{KL}}(Q \parallel P) = \int q(x) \log \frac{q(x)}{p(x)} \, \mathrm{d}x = \mathbb{E}_{x \sim Q}[\log q(x) - \log p(x)].$$

For discrete distributions: $D_{\mathrm{KL}}(Q \parallel P) = \sum_x Q(x) \log \frac{Q(x)}{P(x)}$.

Properties of KL Divergence:

**1. Nonnegative**

$$D_{\mathrm{KL}}(Q \parallel P) \geq 0.$$

Individual log-ratio terms can be negative, but their average cannot.

**2. Zero exactly when distributions match**

$$D_{\mathrm{KL}}(Q \parallel P) = 0 \Leftrightarrow Q = P.$$

Densities may differ on sets of measure zero.

**3. Direction matters**

$$D_{\mathrm{KL}}(Q \parallel P) \neq D_{\mathrm{KL}}(P \parallel Q)$$

in general. KL is **not a distance metric**.

**4. Missing support gives infinite KL**

If $P(A) = 0$ but $Q(A) > 0$, then

$$D_{\mathrm{KL}}(Q \parallel P) = \infty.$$

# Appendix I: Gaussian KL and Regression

Let $Q$ be the **target distribution** and $P_\theta$ the **learned approximation**, with the same covariance:

$$Q = \mathcal{N}(\mu_q, \sigma^2 I), \quad P_\theta = \mathcal{N}(\mu_\theta, \sigma^2 I).$$

$$D_{\mathrm{KL}}(Q \parallel P_\theta) = \frac{\|\mu_q - \mu_\theta\|^2}{2\sigma^2}.$$

### Mean mismatch

Squared error between the target mean $\mu_q$ and predicted mean $\mu_\theta$.

### Variance weighting

Smaller $\sigma^2$ penalizes the same mean error more strongly.

For fixed $\sigma^2$, minimizing KL is equivalent to **minimizing squared error between the means**.

# Appendix II: Why ELBO is a Lower Bound to Data Log-Likelihood



### 1. The noisy states are latent variables

Write the whole noisy trajectory as $z = x_{1:T}$. The generative model defines the joint distribution:

$$p_\theta(x_0, z) = p(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t).$$

The likelihood integrates out all the intermediate states:

$$p_\theta(x_0) = \int p_\theta(x_0, z)\, \mathrm{d}z.$$

The difficulty is the **unknown intermediate distributions**.

### 2. Introduce the known forward process

For a fixed clean image $x_0$, we can easily sample trajectories from $q(z \mid x_0)$. Multiply and divide inside the integral by this density:

$$\begin{aligned} \log p_\theta(x_0) &= \log \int q(z \mid x_0) \frac{p_\theta(x_0, z)}{q(z \mid x_0)}\, \mathrm{d}z \\ &= \log \mathbb{E}_{q(z \mid x_0)} \left[ \frac{p_\theta(x_0, z)}{q(z \mid x_0)} \right]. \end{aligned}$$

So far, this is an **exact equality**.

# Appendix II: Why ELBO Is a Lower Bound

### 3. Apply Jensen's inequality

Replace the **log of an expectation** with the **expectation of a log**.

**Jensen's inequality**
Since log is concave:

$$\log \mathbb{E}[Y] \geq \mathbb{E}[\log Y].$$

$$\log p_\theta(x_0) \geq \underbrace{\mathbb{E}_{q(z \mid x_0)}[\log p_\theta(x_0, z) - \log q(z \mid x_0)]}_{\mathcal{L}_{\text{ELBO}}(x_0)}.$$

### 4. Why does this produce the three terms?

Factor the forward trajectory in reverse order, **conditioning on the known clean image**:

$$q(x_{1:T} \mid x_0) = q(x_T \mid x_0) \prod_{t=2}^{T} q(x_{t-1} \mid x_t, x_0).$$

Substituting this and the generative factorization gives:

$$\mathcal{L}_{\text{ELBO}}(x_0) = \mathbb{E}_{q(x_1 \mid x_0)}[\log p_\theta(x_0 \mid x_1)] - D_{\text{KL}}(q(x_T \mid x_0) \parallel p(x_T))$$
$$- \sum_{t=2}^{T} \mathbb{E}_{q(x_t \mid x_0)}[D_{\text{KL}}(q(x_{t-1} \mid x_t, x_0) \parallel p_\theta(x_{t-1} \mid x_t))].$$

# Appendix III: From the ELBO to Noise Prediction

## 1. Start with a reverse-step KL in the negative ELBO

Maximizing the ELBO means minimizing its negative. For $t \geq 2$, the reverse-step terms are

$$\mathbb{E}_q[D_{\mathrm{KL}}(q(x_{t-1} \mid x_t, x_0) \parallel p_\theta(x_{t-1} \mid x_t))].$$

Both distributions are Gaussian:

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}\left(\tilde{\mu}_t, \tilde{\beta}_t I\right), \quad p_\theta(x_{t-1} \mid x_t) = \mathcal{N}(\mu_\theta, \sigma_t^2 I).$$

Assume the reverse variance $\sigma_t^2$ is **fixed, not learned**. The Gaussian KL becomes

$$D_{\mathrm{KL}}(q \parallel p_\theta) = \frac{1}{2\sigma_t^2}\|\tilde{\mu}_t - \mu_\theta\|^2 + C_t.$$

Here, $C_t$ does not depend on $\theta$. The target and model variances need not be equal.

Thus, training the reverse distribution reduces to **matching its mean**.

# Appendix III: From the ELBO to Noise Prediction

## 2. Substitute the exact and learned means

**Exact posterior mean**

$$\tilde{\mu}_t = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\overline{\alpha}_t}}\varepsilon\right).$$

**Learned reverse mean**

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\overline{\alpha}_t}}\varepsilon_\theta(x_t, t)\right).$$

Subtracting cancels $x_t$:

$$\tilde{\mu}_t - \mu_\theta = \frac{\beta_t}{\sqrt{\alpha_t}\sqrt{1-\overline{\alpha}_t}}(\varepsilon_\theta(x_t, t) - \varepsilon).$$

Squaring gives:

$$\|\tilde{\mu}_t - \mu_\theta\|^2 = \frac{\beta_t^2}{\alpha_t(1-\overline{\alpha}_t)}\|\varepsilon - \varepsilon_\theta(x_t, t)\|^2.$$

$$D_{\mathrm{KL}}(q \parallel p_\theta) = w_t\|\varepsilon - \varepsilon_\theta(x_t, t)\|^2 + C_t, \quad w_t = \frac{\beta_t^2}{2\sigma_t^2\alpha_t(1-\overline{\alpha}_t)}.$$

The Gaussian reverse-step KL is exactly a **timestep-weighted noise-prediction error** plus a constant independent of $\theta$.

# Appendix III: From the ELBO to Noise Prediction

### 3. Sum the reverse-step contributions

Construct the noisy state using

$$x_t = \sqrt{\overline{\alpha}_t} x_0 + \sqrt{1 - \overline{\alpha}_t}\varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I).$$

Up to constants independent of $\theta$, the reverse-step contribution to the negative ELBO is

$$\sum_{t=2}^{T} \mathbb{E}_{x_0,\varepsilon}\left[ w_t \left\| \varepsilon - \varepsilon_\theta\left( \sqrt{\overline{\alpha}_t} x_0 + \sqrt{1 - \overline{\alpha}_t}\varepsilon, t \right) \right\|^2 \right].$$

Instead of evaluating every timestep for each image, sample $t$ uniformly. For any quantities $f_t$,

$$\sum_{t=2}^{T} f_t = (T-1)\mathbb{E}_{t\sim \text{ Uniform}(\{2,\dots,T\})}[f_t].$$

Random timestep sampling **estimates the sum**. It **does not remove** the weights $w_t$.

# Appendix III: From the ELBO to Noise Prediction

### 4. Deliberately remove the timestep weights

Sample $x_0 \sim q_{\text{data}}$, $t \sim \text{Uniform}(\{1, ..., T\})$, and $\varepsilon \sim \mathcal{N}(0, I)$.

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{x_0,t,\varepsilon}\left[\left\|\varepsilon - \varepsilon_\theta\left(\sqrt{\overline{\alpha}_t}x_0 + \sqrt{1-\overline{\alpha}_t}\varepsilon, t\right)\right\|^2\right].$$

This gives each sampled timestep **equal explicit weight**. It changes their relative importance and can change the solution learned by a shared neural network.

**What about the other ELBO terms?**

- **Terminal prior KL:** independent of $\theta$ when the forward process and prior are fixed.
- **Reconstruction at $t = 1$:** a separate likelihood term in the exact ELBO. The simplified objective instead uses noise-prediction MSE at $t = 1$ as well.

# Appendix IV: Deriving the Exact Posterior

We seek the distribution of $x_{t-1}$ given $x_t$ and the clean image $x_0$, for $t \geq 2$.

**1. Apply Bayes' rule**

$$q(x_{t-1} \mid x_t, x_0) = \frac{q(x_t \mid x_{t-1}, x_0) q(x_{t-1} \mid x_0)}{q(x_t \mid x_0)}.$$

The forward process is **Markov**:

$$q(x_t \mid x_{t-1}, x_0) = q(x_t \mid x_{t-1}).$$

The denominator does not depend on the unknown $x_{t-1}$. Therefore,

$$q(x_{t-1} \mid x_t, x_0) \propto q(x_t \mid x_{t-1}) q(x_{t-1} \mid x_0).$$

Both factors are known Gaussians:

$$q(x_t \mid x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t} x_{t-1}, \beta_t I),$$

$$q(x_{t-1} \mid x_0) = \mathcal{N}\left(x_{t-1}; \sqrt{\overline{\alpha}_{t-1}} x_0, (1 - \overline{\alpha}_{t-1}) I\right).$$

# Appendix IV: Multiply and Expand the Gaussians



**2. Write the product using Gaussian exponents**

Let $y = x_{t-1}$. Then

$$q(y \mid x_t, x_0) \propto \exp\left[-\frac{1}{2}\left(\frac{\left\|x_t - \sqrt{\alpha_t}y\right\|^2}{\beta_t} + \frac{\left\|y - \sqrt{\overline{\alpha}_{t-1}}x_0\right\|^2}{1-\overline{\alpha}_{t-1}}\right)\right].$$

Here, $x_t$ and $x_0$ are fixed observations; $y$ **is the variable whose distribution we seek**.

**3. Expand and collect the terms involving** $y$

$$\left\|x_t - \sqrt{\alpha_t}y\right\|^2 = \|x_t\|^2 - 2\sqrt{\alpha_t}x_t^\top y + \alpha_t\|y\|^2.$$

$$\left\|y - \sqrt{\overline{\alpha}_{t-1}}x_0\right\|^2 = \|y\|^2 - 2\sqrt{\overline{\alpha}_{t-1}}x_0^\top y + \overline{\alpha}_{t-1}\|x_0\|^2.$$

Dropping terms independent of $y$:

$$q(y \mid x_t, x_0) \propto \exp\left[-\frac{1}{2}(A\|y\|^2 - 2b^\top y)\right],$$

where

$$A = \frac{\alpha_t}{\beta_t} + \frac{1}{1-\overline{\alpha}_{t-1}}, \quad b = \frac{\sqrt{\alpha_t}}{\beta_t}x_t + \frac{\sqrt{\overline{\alpha}_{t-1}}}{1-\overline{\alpha}_{t-1}}x_0.$$

# Appendix IV: Complete the Square

**4. Complete the square**

Use the identity

$$A\|y\|^2 - 2b^\top y = A\left\|y - \frac{b}{A}\right\|^2 - \frac{\|b\|^2}{A}.$$

The final term is independent of $y$ and is absorbed into the normalizing constant:

$$q(y \mid x_t, x_0) \propto \exp\left[-\frac{A}{2}\left\|y - \frac{b}{A}\right\|^2\right].$$

Compare with the isotropic Gaussian density:

$$\mathcal{N}(y; \mu, vI) \propto \exp\left[-\frac{1}{2v}\|y - \mu\|^2\right].$$

$$\tilde{\mu}_t = \frac{b}{A}, \quad \tilde{\beta}_t = \frac{1}{A}.$$

Thus,

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}\left(x_{t-1}; \tilde{\mu}_t, \tilde{\beta}_t I\right).$$

# Appendix IV: Simplify the Posterior Parameters



**5. Obtain the posterior variance and mean**

Since $\alpha_t + \beta_t = 1$ and $\overline{\alpha}_t = \alpha_t \overline{\alpha}_{t-1}$,

$$A = \frac{\alpha_t(1-\overline{\alpha}_{t-1}) + \beta_t}{\beta_t(1-\overline{\alpha}_{t-1})}$$
$$= \frac{1-\overline{\alpha}_t}{\beta_t(1-\overline{\alpha}_{t-1})}.$$

Hence the **posterior variance** is

$$\tilde{\beta}_t = \frac{1}{A} = \frac{1-\overline{\alpha}_{t-1}}{1-\overline{\alpha}_t}\beta_t.$$

Multiplying $b$ by this variance gives the **posterior mean**:

$$\tilde{\mu}_t = \tilde{\beta}_t \left( \frac{\sqrt{\alpha_t}}{\beta_t} x_t + \frac{\sqrt{\overline{\alpha}_{t-1}}}{1-\overline{\alpha}_{t-1}} x_0 \right).$$

$$\tilde{\mu}_t(x_t, x_0) = \frac{\sqrt{\overline{\alpha}_{t-1}}\beta_t}{1-\overline{\alpha}_t} x_0 + \frac{\sqrt{\alpha_t}(1-\overline{\alpha}_{t-1})}{1-\overline{\alpha}_t} x_t.$$